Lu Zhang | VFX 721 | Spring 2026

Project 03 | Procedural Animation

Houdini 21.0 / Karma XPU

May 26, 2026

Wired Wings - User Guide and Breakdown

Render Statistics

Average render time	1.8 min/frame
Resolution	1080 × 1920
Noise threshold	0.01
Samples per pixel	Min 1 · Max 9

Geometry Statistics

Polygons	~322,004
Curves	~3,120

Lights

Rectangle Light	2
Point Light	1

Project Description

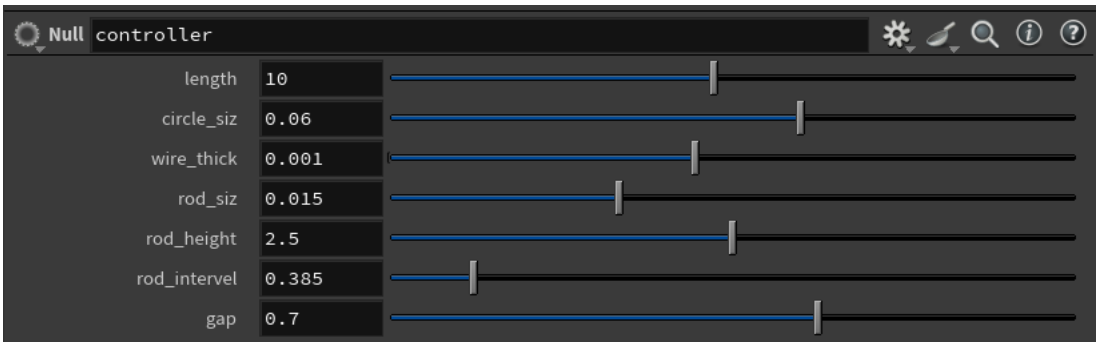
Wired Wings is a procedural animation project recreating a real kinetic installation sculpture exhibited at Parking Gallery, Moscow. The original work features multiple paper wing panels mounted on an intricate steel wire armature with ring joints, pendulum counterweights, and curved wire connectors. This project reconstructs the installation in Houdini and animates it procedurally, simulating the rhythmic flapping motion of the wings as if driven by an invisible mechanical breath.

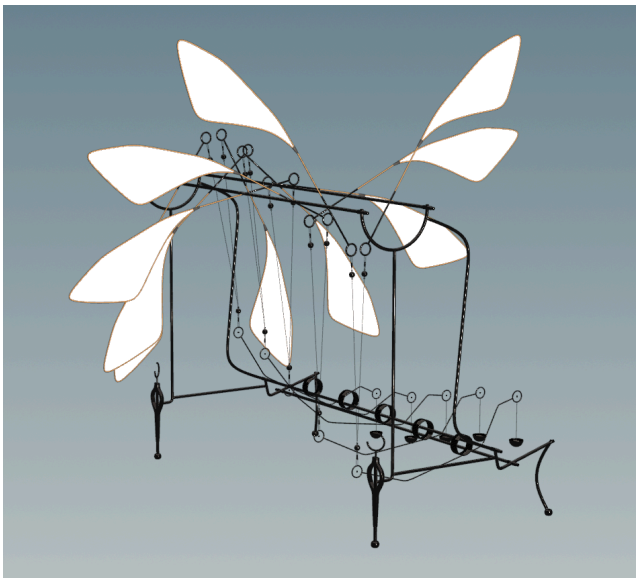


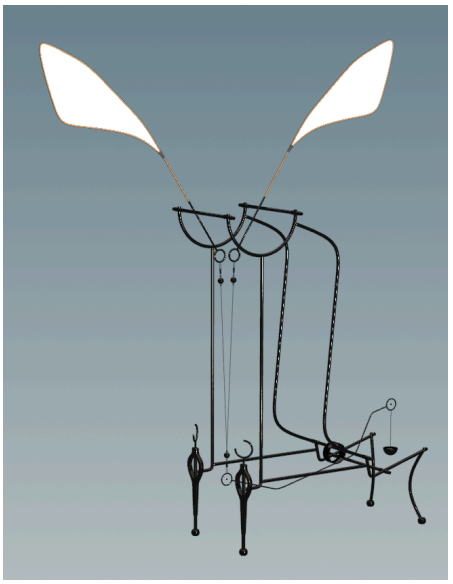
Left: reference Right: render

Technical guide

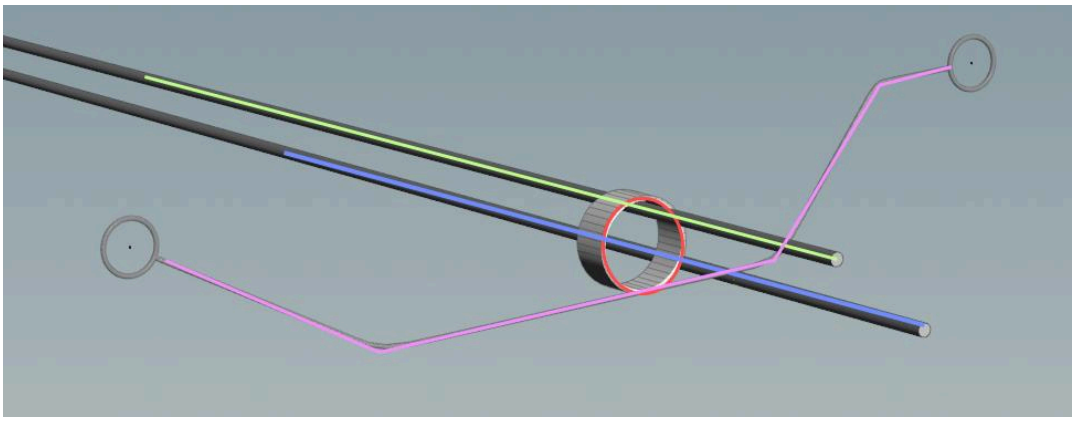
1. Controller

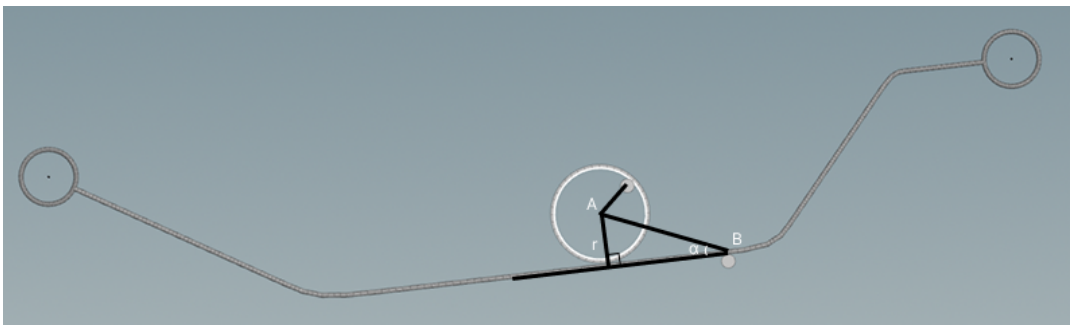






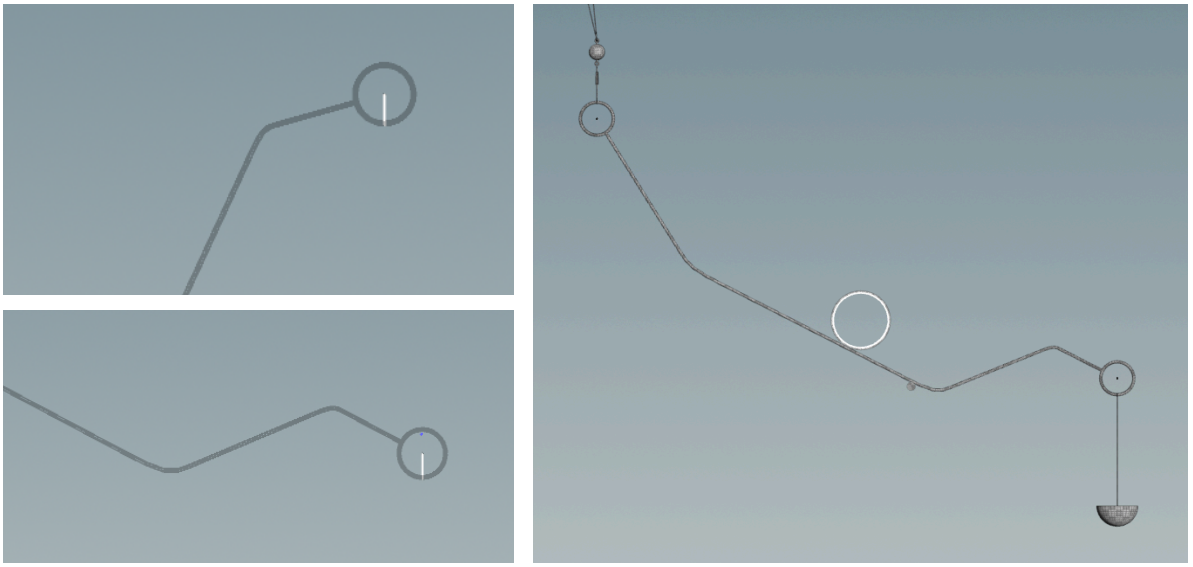
2. Drive Mechanism (Cam-Follower System)





- Cam circle fixed to green shaft, rotates continuously driven by $FF * 2
- Follower arm (pink) pivoted at point B on the fixed horizontal rod
- As shaft rotates, cam center position A is computed first
- Tangent angle solved analytically:
    - atan2 → direction angle from B to A
    - acos(r / |AB|) → offset angle α between BA line and tangent line
    - Direction angle − α − 90° = final rotation of follower arm
- Converts continuous shaft rotation into oscillating swing of follower arm

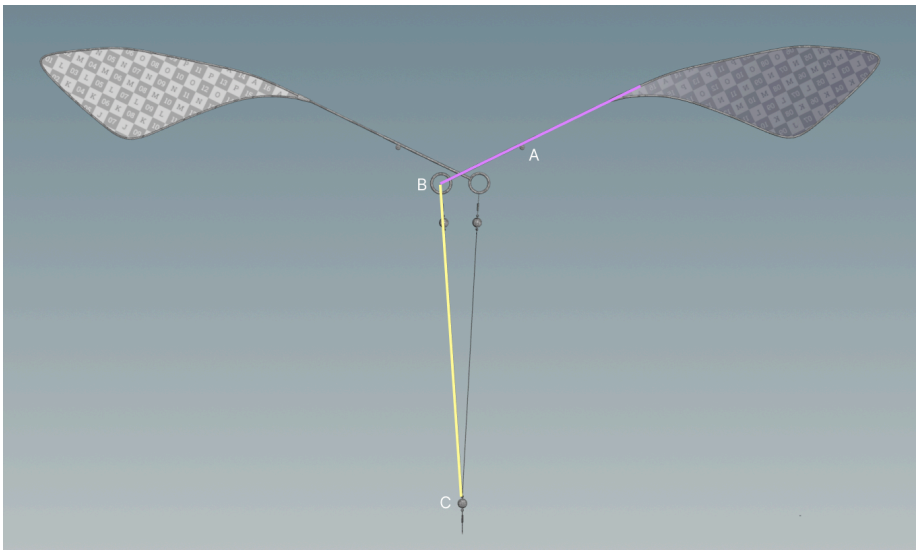
3. Endpoint Extraction



- Ring endpoints located using Intersection Analysis node

- Right-side counterweight: vertical line cast downward from ring center → intersection always yields lowest point on circle
- Left-side hook: same method with differently oriented line → extracts corresponding contact point
- Ensures correct attachment positions regardless of arm rotation

4. Triangle Linkage



- A (wing attachment point) and B (central ring joint) are fixed; C is driven by cam-follower system
- AB = wing support rod; BC = suspension wire; all three side lengths fixed
- Each frame, rotation of AB and CB solved analytically using law of cosines:
- atan2 → base direction angle between endpoints
- acos → angular offset between baseline and arm
- Combined → final world-space rotation of each arm
- Result: wing flaps as C oscillates

BIBLIOGRAPHY AND REFERENCES

- https://yandex.com/maps/org/parking_galereya/36006402022/gallery/?ll=37.628833%2C55.752959&mode=search&photos%5Bbusiness%5D=36006402022&photos%5Bid%5D=vplv4tbikluivcccz6tj&sll=37.624713%2C55.752959&tab=gallery&text=%D0%BF%D0%BE%D0%B4%D0%B7%D0%B5%D0%BC
- https://pin.it/3PWHmXERa